

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

Anonymous WACV [Datasets Track](#) submission

Paper ID *****

Abstract

001 Long-horizon video generation is evaluated with whole-
002 frame metrics that reward motion and temporal consis-
003 tency. For fixed-camera nature scenes this creates an
004 ambiguity: motion of water, fire, smoke, or rain is de-
005 sirable, whereas motion of the background is an error.
006 A system can therefore score well on motion while its
007 scene drifts, or on consistency while its flow stagnates.
008 We introduce SNF-Bench, an evaluation framework for
009 long-horizon fixed-camera generation that partitions
010 each scene into static support and dynamic flow and
011 reports static fidelity, flow persistence with absolute
012 magnitude, and drift leakage separately, never as one
013 score. Each factor is validated mechanistically rather
014 than by correlation with preference: we inject global
015 translation, rotation, and scale drift and progressive
016 motion attenuation at known severity into real gener-
017 ations, and require each factor to respond in its stated
018 direction and to remain selective against corruptions it
019 does not target. Auditing seven publicly released long-
020 horizon generators in their intended configurations, we
021 find that whole-frame motion and static-region drift
022 induce near-opposite orderings of the same outputs,
023 and that the controlled drift our static-fidelity fac-
024 tors detect increases a widely used whole-frame mo-
025 tion score. SNF-Bench measures where motion occurs
026 and whether it persists; it does not measure physical
027 realism.

028 1. Introduction

029 Autoregressive and streaming video diffusion models
030 now extend generation from seconds to minute-scale
031 horizons. Evaluation has not kept pace: small tempo-
032 ral errors invisible in short clips accumulate over long
033 rollouts into background displacement, colour drift,
034 or motion decay, and whole-frame metrics summarise
035 these qualitatively different failures into one score.

The problem is sharpest in fixed-camera nature
scenes. In a static-camera video of a river, the wa-
ter should move while the bank, rocks, and surround-
ing landscape stay fixed. The same separation gov-
erns waterfalls, ocean waves, rainfall, fire, smoke, and
wind-driven vegetation: some regions form static sup-
port that must remain structurally fixed, while others
carry dynamic flow that must persist. Whole-frame
motion measures reward optical flow regardless of its
origin, crediting background drift as readily as genuine
fluid motion, while temporal-consistency measures can
favour videos whose stability comes from suppressing
the flow they should sustain. Two opposite failures—
static drift and flow decay—are thereby conflated, and
increasingly so at long horizons, where accumulated
background motion can dominate a sequence’s appar-
ent dynamics. Figure 1 shows both failures in released
generations of a single prompt.

We introduce SNF-Bench (Static-Natural Flow), an
evaluation framework built around this decomposition,
evaluated at 60 and 120s horizons with shorter and
longer tiers retained as diagnostics. Measurement is
organised into three factors. Static fidelity captures
geometric displacement and residual optical flow inside
nominally stationary regions. Flow persistence reports
the retention of dynamic-region motion together with
its absolute compensated magnitude, so a retention
ratio cannot appear strong merely because motion is
uniformly weak. Drift leakage relates static-region mo-
tion and global-motion compensation to the measured
dynamic-region flow.

A benchmark is only as trustworthy as its measure-
ments, so we validate the protocol independently of
any model comparison. Controlled perturbations in-
ject global translation, rotation, and scale drift, pro-
gressive motion attenuation, late freezing, photometric
drift, and temporal repetition into real fixed-camera
generations at known severity, and we require each fac-
tor to respond in the direction its definition states and
to remain selective against corruptions it does not tar-

036
037
038
039
040
041
042
043
044
045
046
047
048
049
050
051
052
053
054
055
056
057
058
059
060
061
062
063
064
065
066
067
068
069
070
071
072
073
074
075



Figure 1. Why a fixed-camera benchmark needs a spatial decomposition. Three released 60s generations of one prompt. A whole-frame motion score rates the first two rows almost identically, yet only the first holds its scene in place; their static-region drift differs by over an order of magnitude. The third row shows the opposite failure, which that score does detect.

get. Because a parameter expressed in degrees is not commensurate with one expressed in pixels, geometric severity is specified as the mean displacement a corruption induces, which makes the families directly comparable. We additionally recompute every measurement under systematic erosion and dilation of the region partition.

We then audit seven publicly released long-horizon generators in their intended configurations. The audit exposes cases where conventional metrics and SNF-Bench interpret the same outputs differently: high whole-frame motion coexisting with substantial static-region displacement, and high stability coexisting with decaying flow. We do not argue that existing metrics are incorrect. They answer a whole-frame question accurately; that question does not encode the spatial distinction this setting requires. Nor does SNF-Bench measure physical realism—a sequence may be drift-free and persistent yet physically implausible. The contribution is narrower and complementary: making the spatial origin and temporal persistence of motion explicit.

Our contributions are:

1. SNF-Bench, an evaluation framework for long-horizon fixed-camera generation organised around a static-support/dynamic-flow partition.
2. A measurement protocol that reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately, and never as a single score.
3. Mechanistic validation of every reported factor under corruptions of known type and severity, including calibration of the drift diagnostics and robustness of the region partition.
4. A public-model audit showing that this decomposition materially changes the interpretation and ordering of long-horizon behaviour relative to whole-frame metrics.

2. Related Work

Video-generation evaluation. General-purpose suites assess complementary aspects of visual quality, temporal consistency, motion, and semantic alignment. VBench [8] decomposes quality into dimensions including background consistency, motion smoothness, temporal flickering, and dynamic degree, and VBench-2.0 [25] extends the suite toward intrinsic faithfulness; EvalCrafter [13] learns a weighting of many metrics from user opinion; Video-Bench [4] scores generations with multimodal language models validated against human preference. Others specialise: T2V-CompBench [19] targets compositional binding, VM-Bench [11] targets perceptually aligned motion quality, and WorldScore [1] evaluates controllability, quality and dynamics for world generation. These suites set the standards a benchmark paper is expected to meet—an explicit taxonomy, validated metrics, stated scale—but their measurements are whole-frame or sequence-level, and a whole-frame aggregate cannot reveal whether motion originates where it was intended or in displacement of nominally static structure. VM-Bench is the closest comparator, since it too reinterprets motion, but it asks whether motion is perceptually plausible where we ask where in the frame it occurs and whether it survives compensation for global displacement. Our validation is correspondingly mechanistic—perturb by a known amount, require the expected response—complementing rather than replacing the human-alignment evidence those suites provide.

Long-horizon autoregressive generation. Autoregressive and streaming generators extend synthesis beyond short bidirectional clips by producing frames or temporal chunks causally, including CausVid [24], Self-Forcing [7], Infinite-Forcing [9], Rolling-Forcing [12],

Reward-Forcing [14], LongLive [22], and Causal-Forcing [26]; bidirectional systems such as LTX-Video [3] and Wan [21] provide short-horizon reference points. These systems differ substantially in temporal block size, training objective, memory and cache policy, and horizon-extension mechanism [23], yet their evaluation typically relies on generic video metrics or method-specific qualitative analysis. That diversity motivates auditing each system in its released configuration, and the evaluation gap motivates a system-agnostic question: does static support remain fixed while dynamic flow persists?

Image dynamics and nature-flow animation. Work on animating still images has long modelled natural motion explicitly: controllable fluid animation separates static image structure from specified fluid motion [15], and Generative Image Dynamics models long-term scene motion through learned trajectory representations [10]. These works treat static support and dynamic content as distinct modelling targets; SNF-Bench elevates that distinction to an evaluation principle for long-horizon generators.

3. SNF-Bench Design

SNF-Bench evaluates long-horizon generation under a deliberately constrained setting: a fixed camera observing a scene containing both stationary structure and persistent natural motion. This isolates two failure modes that are difficult to separate in unconstrained video—static drift, where nominally stationary structure moves over time, and flow decay, where intended dynamics weaken or disappear. Every sequence is treated as a composition of two spatial roles, Ω_{static} for regions expected to remain geometrically stationary and Ω_{flow} for regions expected to exhibit sustained motion. The partition defines an evaluation role, not a semantic ontology: a riverbank is static support while the river surface is flow; buildings are static support in a rainfall scene while the falling rain is flow. SNF-Bench does not require dynamics inside Ω_{flow} to follow any physical model. It measures whether motion stays localised to the appropriate regions and persists over time without contaminating static support.

Every system receives identical conditioning per item, and no prompt is rewritten for a particular method. Both conditioning modalities are evaluated under the same decomposition, differing only in how static support is established: the text-conditioned setting builds it in the opening of the sequence, while the image-conditioned setting takes it from a shared source image, so divergence there is attributable to the system rather than to a differently imagined scene. The

main text reports the text-conditioned track for space; complete image-conditioned results appear in the supplementary material.

3.1. Scene Categories and Horizons

Prompts are stratified across representative fixed-camera natural processes—channel water, ocean waves, precipitation, fire and smoke, lava, and wind-borne motion—with the category set fixed before the final sweep. The strata are not equally populated: channel water accounts for 61% of the principal prompt set, which is why every reported aggregate is macro-averaged over the 6 categories rather than averaged over prompts, so that no single medium determines a headline value. Complete per-category counts are given in the supplementary material.

Four temporal regimes are used: 5s as an initialisation check, 60s as the reported horizon, and 120 and 240s as diagnostic tiers. Conclusions rest on the 60s tier, which carries 23 prompts; the longer tiers are reported in the supplementary material, since their samples do not support method-level separation on their own. A system that cannot reach a horizon under its released configuration is reported as unsupported rather than silently modified.

3.2. Region Partition

The partition is obtained automatically and identically for every sequence. Mean optical-flow magnitude over the leading 12% of sampled frame pairs is thresholded by Otsu’s method [16] into a dynamic region and its complement; the static region is then eroded and a 4% frame border discarded, which together form an ignored transition band so that boundary pixels—where the partition is least certain and flow least reliable—contribute to neither region.

Restricting the estimate to an early window is what prevents the measurement from becoming circular. Deriving a partition from a sequence’s entire motion field would allow accumulated background drift to be read as evidence that the affected region is dynamic; at the 60s horizon the partition is instead fixed from the first 7.1s, before long-horizon drift has accumulated, and the metrics are evaluated over the remainder. The partition therefore reflects each system’s spatial layout, which is unavoidable when every system authors its own scene, but it is not defined by the temporal failure being measured.

Two limitations follow, and both bound the conclusions drawn later. The partition is automatic rather than annotated, so a system that drifts within the first 12% of its rollout can influence its own partition; we therefore treat partitions as measured quantities and

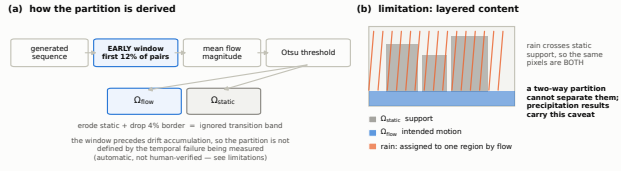


Figure 2. Region-partition protocol. (a) The partition is derived automatically from early-window flow, with an ignored transition band at the boundary. (b) A two-way partition cannot represent layered content such as rain crossing static support.

recompute every result under systematic erosion and dilation of the boundary (Sec. 5). And a two-way partition cannot represent layered content: rain and snow transit static support, so those pixels are assigned to one region by flow magnitude alone and static-fidelity values in precipitation scenes carry an additional unmodelled contribution. Derivation details, the sensitivity study, and the annotation protocol we regard as the natural strengthening of this component are given in the supplementary material.

3.3. Benchmark Manifest

SNF-Bench is defined by a versioned manifest fixed before the final sweep, recording per item: identifier and category, exact prompt, target horizons, resolution and frame-rate metadata, and provenance. After freezing, items are never removed because a system performs poorly; corrections require a new benchmark version. A failure taxonomy—unsupported horizon, out-of-memory, runtime crash, numerical corruption, invalid output—is likewise fixed in advance, and failures are reported alongside successful generations rather than silently excluded. Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial.

4. Drift–Motion Evaluation Protocol

SNF-Bench evaluates fixed-camera generation along three complementary factors: Static Fidelity, Flow Persistence, and Drift Leakage. No individual score is sufficient: low background motion can be achieved by freezing the entire video, and high foreground motion can be produced by drifting the entire scene. We therefore always report static-region stability jointly with both the relative persistence and absolute magnitude of dynamic-region motion.

Let I_t denote frame t of a T -frame sequence, Ω_{static} and Ω_{flow} the fixed region masks of Sec. 3.2, and $\mathbf{u}_t(\mathbf{x})$ the optical flow from I_t to I_{t+1} at pixel $\mathbf{x}$, estimated

with RAFT [20]. Each sequence defines an early window $\mathcal{E}$ and a late window $\mathcal{L}$ of equal duration; $\mathcal{E}$ begins after a short warm-up interval so that initial generation transients do not define the motion reference. Window extents are fixed in the benchmark specification.

4.1. Static Fidelity

Feature-aligned Background Drift (fBD). We extract ORB features [18] inside Ω_{static} from the first frame and match them against each frame I_t in the late window, applying robust geometric filtering to reject outlier correspondences. For the resulting matched pairs $\mathcal{M}_t$,

$$D_{feat}(t) = \text{median}_{(p,q) \in \mathcal{M}_t} \|p - q\|_2, \quad (1)$$

$$\text{fBD} = \frac{100}{d} \cdot \frac{1}{|\mathcal{L}|} \sum_{t \in \mathcal{L}} D_{feat}(t) \quad (\downarrow),$$

where d is the frame diagonal, so fBD is expressed as a percentage of image size. fBD captures accumulated geometric displacement of persistent structure and is intentionally insensitive to fine-scale flicker that moves no structure.

Normalized Background Flow (NBF). Complementing accumulated drift, NBF measures instantaneous motion energy inside the static mask:

$$\text{NBF} = \frac{10^3}{W \Delta t} \cdot \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega_{static}|} \sum_{\mathbf{x} \in \Omega_{static}} \|\mathbf{u}_t(\mathbf{x})\|_2 \quad (\downarrow), \quad (2)$$

normalized by frame width W for resolution invariance and by the inter-frame interval Δt for frame-rate invariance, giving units of 10^{-3} frame widths per second. Here Δt is the interval of the sampled pairs actually used for flow estimation: methods are evaluated at their native frame rates, which differ across the roster, and the evaluation pipeline subsamples each clip to a common temporal rate before measurement. Expressing NBF per unit time rather than per frame makes the quantity independent of both the native rate and the sampling policy, so the measure remains comparable if either changes.

NBF is a normalized magnitude, not a ratio: we deliberately do not divide by whole-frame flow, since that would couple background leakage to foreground behavior and inflate the measure for methods whose dynamic motion decays.

4.2. Flow Persistence

Global-motion compensation. Coherent scene displacement is removed before dynamic-region motion is evaluated. A single translation vector is insufficient:

under rotation or zoom the static-region flow field is not constant, and its component-wise median is near zero for a centered rotation, so a translation-only estimate would leave exactly the corruptions Sec. 5 injects uncompensated. We therefore estimate a robust global similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ from static-region correspondences $\mathbf{x} \mapsto \mathbf{x} + \mathbf{u}_t(\mathbf{x})$ under RANSAC, define the induced global displacement field $\mathbf{d}_t(\mathbf{x}) = T_t(\mathbf{x}) - \mathbf{x}$, and compensate pointwise:

$$\tilde{\mathbf{u}}_t(\mathbf{x}) = \mathbf{u}_t(\mathbf{x}) - \mathbf{d}_t(\mathbf{x}). \quad (3)$$

We restrict the model to a similarity rather than an unrestricted affine transform because affine shear can absorb legitimate local deformation and over-correct. Where correspondences are insufficient or the robust fit fails we fall back to median translation and record that fallback per sequence.

Motion-Compensated Foreground Flow (MCFF). For a temporal window $\mathcal{W}$,

$$\text{MCFF}(\mathcal{W}) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \frac{1}{|\Omega_{\text{flow}}|} \sum_{\mathbf{x} \in \Omega_{\text{flow}}} \|\tilde{\mathbf{u}}_t(\mathbf{x})\|_2 \quad (\uparrow). \quad (4)$$

We report $\text{MCFF}(\mathcal{E})$ and $\text{MCFF}(\mathcal{L})$, so a method with uniformly negligible motion cannot appear strong through relative measures alone.

Flow Persistence (FP).

$$\text{FP} = \min\left(\frac{\text{MCFF}(\mathcal{L})}{\text{MCFF}(\mathcal{E}) + \epsilon}, c\right) \quad (\uparrow), \quad (5)$$

clipped at a fixed constant c to prevent unstable ratios when early motion is small. FP near one indicates retained motion; substantially smaller values indicate decay; values above one are not automatically better. FP is never reported without its MCFF magnitudes.

4.3. Drift Leakage and Attenuation

Whole-frame metrics cannot determine whether apparent dynamic-region motion arises from intended flow or from displacement of the whole scene. We report two diagnostics, since neither subsumes the other. Let F_{static} denote mean flow magnitude over Ω_{static} , and $F_{\text{raw}}, F_{\text{comp}}$ the mean raw and compensated flow magnitude over Ω_{flow} , all in the late window; the late window is used because drift accumulates over the rollout.

Drift Leakage Ratio (DLR).

$$\text{DLR} = \frac{F_{\text{static}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow) \quad (6)$$

locates the sequence’s motion energy. It is not a fraction and is deliberately unbounded: $\text{DLR} > 1$ indicates that static support moves more than the region intended to move, a regime that occurs in practice.

Drift Attenuation Ratio (DAR).

$$\text{DAR} = 1 - \frac{F_{\text{comp}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow) \quad (7)$$

measures the signed relative change in dynamic-region flow after compensation—how much apparent motion a global displacement model accounts for. Reported values are clipped to $[0, 1]$ while the signed quantity is retained for validation; negative values, which arise where local flow opposes the estimated global field, are reported rather than suppressed.

Interpretation. Neither quantity is a causal decomposition of motion into drift and non-drift components, and we do not describe them as one. Read together they are more informative than either alone: high DLR with moderate DAR indicates large static-region motion energy that a rigid global model does not explain, implicating non-rigid instability such as warping rather than uniform scene translation.

Semantic guardrail. The decomposition is agnostic to what moves: a sequence rendering drifting fog where a river was requested can occupy a favourable operating point. We therefore report one established prompt-alignment score alongside the factors—CLIP-based image-text similarity [6] averaged over sampled frames—as a contextual column, not a fourth factor, and never aggregated with the others. Its role is to make a semantically wrong generation visible so the spatial factors are read only where the requested phenomenon is present. Learned video-quality predictors [5] serve the same purpose at higher cost and with heavier dependence on their own training distribution.

4.4. Joint Interpretation

No factor is read alone. Low static drift together with low compensated motion indicates an over-stabilised, nearly frozen sequence; high compensated motion together with high static-region flow indicates motion contaminated by displacement; high retention with negligible magnitude indicates persistent but trivial motion. The desired regime is low static drift, non-trivial compensated motion, and high retention. SNF-Bench therefore reports no single scalar score: collapsing the factors would recreate the ambiguity it exists to expose.

414 Implementation convention. Videos are normalised to
415 a common evaluation resolution and temporally sub-
416 sampled to a common flow-estimation rate; no frame
417 interpolation is performed, since synthesising inter-
418 mediate frames would alter the flow field being mea-
419 sured. Every estimator, threshold, window extent, and
420 constant is fixed globally before the sweep and ap-
421 plied unchanged to every system; the full list, and the
422 per-sequence frame rate, resolution and compensation
423 mode released alongside the scores, are given in the
424 supplementary material.

425 5. Controlled Validation

426 Before SNF-Bench is used to compare generators, we
427 validate that its measurements respond to the failure
428 modes they target. A benchmark is not justified merely
429 because its metrics produce different rankings; each
430 metric must change selectively, monotonically, and in
431 the expected direction under known corruptions. We
432 therefore construct controlled variants of fixed-camera
433 reference clips with known corruption type and sever-
434 ity. Validation is performed independently of the audit
435 in Sec. 6: perturbation strengths, masks, temporal win-
436 dows, and metric hyperparameters are fixed before the
437 final baseline results are inspected.

438 5.1. Perturbations and Criteria

439 Given a reference sequence we generate variants
440 that isolate individual components of the decom-
441 position. Geometric drift applies a temporally in-
442 creasing global transformation—translation, rotation,
443 or scale—simulating camera-like background displace-
444 ment that accumulates as long-horizon error does. Mo-
445 tion attenuation progressively blends dynamic-region
446 content toward an early reference, and late freezing
447 replaces all content after a chosen fraction of the se-
448 quence, both leaving static support untouched. Photo-
449 metric drift varies brightness and colour temperature
450 while preserving geometry and motion, bounding the
451 residual sensitivity of flow-based quantities. Temporal
452 repetition replaces late dynamic content with a short
453 repeated cycle. Partition perturbation erodes and di-
454 lates the region boundary rather than the video.

455 Geometric severity is specified as the mean pixel
456 displacement a corruption induces, with each family’s
457 parameter derived from it using the frame geometry.
458 This is not presentational: a parameter in degrees is
459 not commensurate with one in pixels, and at the eval-
460 uation resolution one degree of rotation induces only
461 a few pixels of displacement, so families compared by
462 their native parameters differ in injected severity by
463 more than an order of magnitude.

464 A factor is admitted to the audit only if its response

direction matches its definition, that response is mono-
tonic in severity, unrelated corruptions produce sub-
stantially smaller changes, and conclusions survive per-
turbation of the region partition. Factors failing these
checks are revised or reported as supplementary diag-
nostics rather than headline quantities. Complete per-
turbation levels, the calibration procedure for the drift
diagnostics, and the full response matrix are given in
the supplementary material.

Complementary checks. Mechanistic validation shows
a factor responds to the corruption it names, not that
its ordering matches what an observer perceives. Two
narrow checks address that without importing a prefer-
ence score: an axis-specific human study and an open-
weight vision–language judge [17] put to identical ques-
tions, both reported as per-axis agreement and neither
aggregated into a factor (supplementary material).

5.2. Validation Results

Figure 3 and Table 1 report the outcome. Injected
translation produces the expected monotone increase
in both static-fidelity factors, and progressive atten-
uation of dynamic-region content drives MCFF-L and
FP down while leaving the static factors comparatively
unchanged—the selectivity the decomposition requires.
The comparison that motivates the benchmark appears
in the same experiment: the identical injected back-
ground drift that our static-fidelity factors register also
raises VBench Dynamic Degree, so a whole-frame mo-
tion score improves on precisely the corruption a fixed-
camera evaluation should penalise.

Two results are reported because they qualify the
framework rather than support it. fBD responds
strongly and monotonically to translation but behaves
erratically under pure rotation, where feature corre-
spondences degrade; NBF carries the static-fidelity sig-
nal in that regime, which is why the two are reported
jointly and why neither is used alone. Reported along-
side them is the scope boundary of Sec. 5.1: looping
dynamic content sustains both persistence and mag-
nitude, so SNF-Bench cannot distinguish genuine pro-
gression from repetition, and does not claim to.

6. Public-Model Audit

We apply SNF-Bench to publicly released long-horizon
generators. The goal is not a universal leaderboard
but a determination of how deployed systems occupy
the static-fidelity/flow-persistence space when used as
their authors intended.

Systems and setting. The audit covers 7 publicly re-
leased autoregressive or streaming generators with ac-

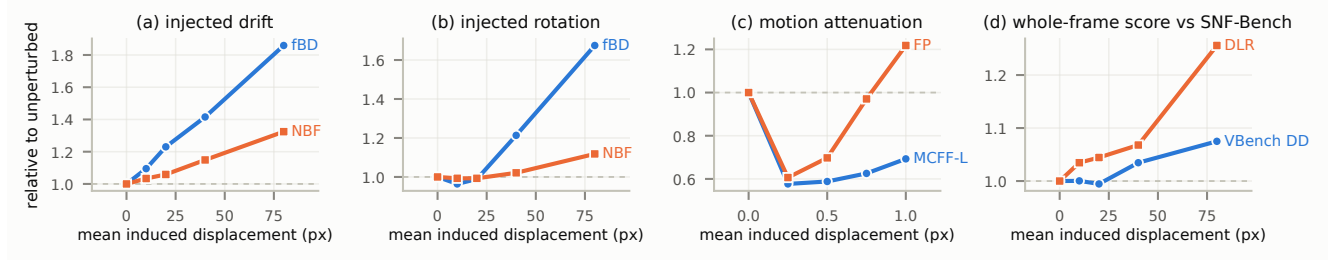


Figure 3. Mechanistic validation. Each panel injects a corruption of known kind and severity into real fixed-camera clips and plots each factor relative to its unperturbed value. Panel (f) puts Vbench Dynamic Degree through the identical corruption as the SNF factors: the score rises with injected background drift. The perturbation is synthetic, so the comparison implicates no published method.

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
Vbench DD	drift	+0.70	+0.70	2%	✓

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25%. Twelve reference clips per severity level.

schedule to populate a table entry.

The image-conditioned track is audited under the same protocol and reported in full in the supplementary material; there, systems evaluated under a common long-horizon wrapper rather than their released configuration are labelled as such, and their scores are read as deployment sensitivity rather than as the published method’s performance.

Controls and statistics. Within each benchmark item every system receives identical conditioning, and the seed set is fixed before the sweep. Seed control guarantees reproducibility within a system; because implementations consume randomness differently, identical seeds do not induce matched noise draws across systems, so cross-system comparison rests on the paired-item design rather than on paired noise. Generated videos are retained in native form, and the normalisation of Sec. 4 is applied unchanged to every system.

Scores are computed per item and macro-averaged across categories. Uncertainty is estimated by percentile bootstrap [2] over items, and we report 95% intervals. Exact ordinal ranks are avoided where uncertainty does not support them: systems whose intervals overlap substantially are not described as separated. Operational limits—unsupported horizons, failed generations, memory exhaustion, invalid outputs—are reported as rates against the frozen manifest and never removed after inspection.

To test whether the decomposition changes interpretation, the same outputs are additionally scored with established whole-frame metrics and we compare the orderings they induce. The focus is interpretation change, not a claim that existing metrics are incorrect.

7. Results and Interpretation Changes

Unless stated otherwise, comparisons refer to systems in their released configurations; deployment-sensitivity

533

534

535

536

537

538

539

540

541

542

543

544

545

546

547

548

549

550

551

552

553

554

555

556

557

558

559

560

561

562

563

564

565

566

567

568

Method	fBD↓	NBF↓	MCFF-L↑	FP↑	DLR↓
60s horizon ($n = 23$)					
CausVid	8.08 ± 3.15	9.27 ± 4.14	1.96 ± 1.33	0.67 ± 0.18	0.45 ± 0.13
Self-Forcing	11.4 ± 3.63	12.7 ± 3.97	2.54 ± 1.40	0.44 ± 0.22	0.50 ± 0.19
Infinite-Forcing	7.75 ± 3.84	3.65 ± 0.88	0.29 ± 0.09	0.63 ± 0.22	0.55 ± 0.11
Rolling-Forcing	16.8 ± 3.69	11.2 ± 2.70	1.57 ± 0.88	0.75 ± 0.29	0.54 ± 0.12
Reward-Forcing	14.8 ± 3.71	8.51 ± 2.79	0.80 ± 0.30	0.49 ± 0.18	0.60 ± 0.13
LongLive	15.9 ± 2.89	13.9 ± 4.76	2.41 ± 1.98	0.75 ± 0.28	0.59 ± 0.13
Causal-Forcing	22.4 ± 2.26	87.6 ± 18.6	7.21 ± 4.17	0.83 ± 0.31	0.94 ± 0.20

Table 2. SNF-Bench audit, T2V track, native configuration. All systems are public external models run under their own intended configuration. Our own systems are excluded by construction. DAR is reported clipped to $[0, 1]$; signed values are released. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel. Values are prompt-level means $\pm$ half the width of a percentile bootstrap 95% interval (10k resamples, fixed seed). MCFF-E and DAR appear in the supplementary table.

results are reported separately in the supplementary material and never redefine a system’s native behaviour.

7.1. Native-Capability Audit

Table 2 reports SNF-Bench measurements at the 60s horizon, the tier whose sample supports method-level separation; all four horizons appear in the supplementary material.

Motion persistence. Dynamic-region behavior is evaluated at the joint operating point (MCFF($\mathcal{L}$), FP), never FP alone: sustained flow requires non-trivial late motion and high retention. Reporting MCFF-E beside MCFF-L makes the degenerate case visible—a system with negligible motion throughout cannot reach a high FP without also displaying a small MCFF-E.

Static–dynamic operating regimes. Figure 4 places each system in the static-drift–surviving-flow plane, exposing qualitatively different regimes—low-drift/low-motion, high-motion/high-drift, and low-drift/non-trivial-motion—without collapsing the axes into a scalar score. The two extremes are informative in opposite directions: Infinite-Forcing attains the lowest static drift of any audited system while placing last on surviving late motion, and Causal-Forcing sits alone at the opposite corner.

7.2. Drift Leakage Changes the Interpretation of Apparent Motion

The effect is systematic rather than anecdotal. Over the audited systems at 60s, VBench Dynamic Degree correlates with SNF-Bench’s static-region flow at

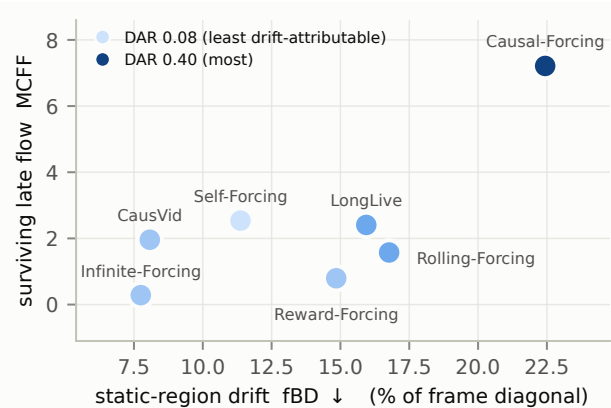


Figure 4. Operating regimes under native deployment. Colour encodes the share of dynamic-region flow a global similarity model accounts for. The view exposes trade-offs; it is not a leaderboard.

Spearman $\rho = +0.93$ (supplementary material): the systems a generic metric reads as most dynamic are, to a close approximation, the systems whose backgrounds move most. The consequence for ordering is direct. Causal-Forcing ranks 1 of 7 on Dynamic Degree and 7 of 7 on NBF; Infinite-Forcing occupies the mirror position, ranking 1 on static fidelity and 7 on apparent motion. Read jointly with its magnitudes, Causal-Forcing’s DLR of 0.94 indicates static-region motion energy almost equal to that of the region intended to move, while its DAR of 0.40 indicates that a rigid global model accounts for only part of it—the remainder is non-rigid instability, which a translation-only account would have misdescribed.

The supplementary material records every case in which the two orderings disagree by more than a fixed rank threshold, under a rule applied to every audited system so that no qualifying case can be omitted.

8. Limitations and Conclusion

SNF-Bench is intentionally narrow: long-horizon fixed-camera generation, where static support and dynamic flow can be meaningfully separated. It does not measure physical realism, and persistence cannot distinguish genuine progression from repetition. The partition is derived automatically rather than from annotation, so a system that drifts early can influence its own partition, and layered content cannot be represented by a two-way split; the transition band and the erosion–dilation study bound this without removing it. Conclusions unstable under partition perturbation are not reported as headline claims, and all measurements inherit the failure modes of the underlying estimators.

References

- [1] Haoyi Duan, Hong-Xing Yu, Sirui Chen, Li Fei-Fei, and Jiajun Wu. WorldScore: A unified evaluation benchmark for world generation. In ICCV, pages 27713–27724, 2025. 2
- [2] Bradley Efron. Bootstrap methods: Another look at the jackknife. *Ann. Statist.*, 7(1):1–26, 1979. 7
- [3] Yoav HaCohen, Nisan Chiprut, Benny Brazowski, Daniel Shalem, Dudu Moshe, Eitan Richardson, Eran Levin, Guy Shiran, Nir Zabari, Ori Gordon, Poriya Panet, Sapir Weissbuch, Victor Kulikov, Yaki Bitterman, Zeev Melumian, and Ofir Bibi. LTX-Video: Realtime video latent diffusion. *arXiv preprint arXiv:2501.00103*, 2024. 3
- [4] Hui Han, Siyuan Li, Jiaqi Chen, Yiwen Yuan, Yuling Wu, Yufan Deng, Chak Tou Leong, Hanwen Du, Junchen Fu, Youhua Li, Jie Zhang, Chi Zhang, Lijia Li, and Yongxin Ni. Video-Bench: Human-aligned video generation benchmark. In CVPR, pages 18858–18868, 2025. 2
- [5] Xuan He, Dongfu Jiang, Ge Zhang, Max Ku, Achint Soni, Sherman Siu, Haonan Chen, Abhranil Chandra, Ziyang Jiang, Aaran Arulraj, Kai Wang, Quy Duc Do, Yuansheng Ni, Bohan Lyu, Yaswanth Narsupalli, Rongqi Fan, Zhiheng Lyu, Bill Yuchen Lin, and Wenhui Chen. VideoScore: Building automatic metrics to simulate fine-grained human feedback for video generation. In *Proc. Conf. Empirical Methods Natural Language Process. (EMNLP)*, pages 2105–2123, 2024. 5
- [6] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. CLIPScore: A reference-free evaluation metric for image captioning. In *Proc. Conf. Empirical Methods Natural Language Process. (EMNLP)*, pages 7514–7528, 2021. 5
- [7] Xun Huang, Zhengqi Li, Guande He, Mingyuan Zhou, and Eli Shechtman. Self forcing: Bridging the train-test gap in autoregressive video diffusion. In *NeurIPS*, 2025. 2
- [8] Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, Yaohui Wang, Xinyuan Chen, Limin Wang, Dahua Lin, Yu Qiao, and Ziwei Liu. VBench: Comprehensive benchmark suite for video generative models. In CVPR, pages 21807–21818, 2024. 2
- [9] Xianglong He Junyi Chen, Zhoujie Fu. Infinite-forcing: Towards infinite-long video generation, 2025. <https://github.com/SOTAMak1r/Infinite-Forcing>. 2
- [10] Zhengqi Li, Richard Tucker, Noah Snavely, and Aleksander Holynski. Generative image dynamics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24142–24153, 2024. 3
- [11] Xinran Ling, Chen Zhu, Meiqi Wu, Hangyu Li, Xiaokun Feng, Cundian Yang, Aiming Hao, Jiashu Zhu, Jiahong Wu, and Xiangxiang Chu. VMBench: A benchmark for perception-aligned video motion generation. In ICCV, 2025. 2
- [12] Kunhao Liu, Wenbo Hu, Jiale Xu, Ying Shan, and Shijian Lu. Rolling forcing: Autoregressive long video diffusion in real time. In ICLR, 2026. 2
- [13] Yaofang Liu, Xiaodong Cun, Xuebo Liu, Xintao Wang, Yong Zhang, Haoxin Chen, Yang Liu, Tiejong Zeng, Raymond Chan, and Ying Shan. Evalcrafter: Benchmarking and evaluating large video generation models. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22139–22149, 2024. 2
- [14] Yunhong Lu, Yanhong Zeng, Haobo Li, Hao Ouyang, Qiuyu Wang, Ka Leong Cheng, Jiapeng Zhu, Hengyuan Cao, Zhipeng Zhang, Xing Zhu, et al. Reward forcing: Efficient streaming video generation with rewarded distribution matching distillation. *arXiv preprint arXiv:2512.04678*, 2025. 3
- [15] Aniruddha Mahapatra and Kuldeep Kulkarni. Controllable animation of fluid elements in still images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3667–3676, 2022. 3
- [16] Nobuyuki Otsu. A threshold selection method from gray-level histograms. *IEEE Trans. Syst. Man Cybern.*, 9(1):62–66, 1979. 3
- [17] Qwen Team. Qwen3-VL technical report. *arXiv preprint arXiv:2511.21631*, 2025. 6
- [18] Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. ORB: An efficient alternative to SIFT or SURF. In ICCV, pages 2564–2571, 2011. 4
- [19] Kaiyue Sun, Kaiyi Huang, Xian Liu, Yue Wu, Zihan Xu, Zhenguo Li, and Xihui Liu. T2V-CompBench: A comprehensive benchmark for compositional text-to-video generation. In CVPR, pages 8406–8416, 2025. 2
- [20] Zachary Teed and Jia Deng. RAFT: Recurrent all-pairs field transforms for optical flow. In ECCV, pages 402–419, 2020. 4
- [21] Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwu Yu, Haiming Zhao, Jianxiao Yang, Jianyuan Zeng, Jiayu Wang, Jingfeng Zhang, Jingren Zhou, Jinkai Wang, Jixuan Chen, Kai Zhu, Kang Zhao, Keyu Yan, Lianghai Huang, Mengyang Feng, Ningyi Zhang, Pandeng Li, Pingyu Wu, Ruihang Chu, Ruili Feng, Shiwei Zhang, Siyang Sun, Tao Fang, Tianxing Wang, Tianyi Gui, Tingyu Weng, Tong Shen, Wei Lin, Wei Wang, Wei Wang, Wenmeng Zhou, Wenten Wang, Wenting Shen, Wenyuan Yu, Xianzhong Shi, Xiaoming Huang, Xin Xu, Yan Kou, Yangyu Lv, Yifei Li, Yijing Liu, Yiming Wang, Yingya Zhang, Yitong Huang, Yong Li, You Wu, Yu Liu, Yulin Pan, Yun Zheng, Yuntao Hong, Yupeng Shi, Yutong Feng, Zeyinzi Jiang, Zhen Han, Zhi-Fan Wu, and Ziyu Liu. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025. 3

- 745 [22] Shuai Yang, Wei Huang, Ruihang Chu, Yicheng Xiao,
746 Yuyang Zhao, Xianbang Wang, Muyang Li, Enze Xie,
747 Yingcong Chen, Yao Lu, Song Han, and Yukang Chen.
748 Longlive: Real-time interactive long video generation.
749 In ICLR, 2026. 3
- 750 [23] Hidir Yesiltepe, Tuna Han Salih Meral, Adil Kaan
751 Akan, Kaan Oktay, and Pinar Yanardag. Infinity-
752 RoPE: Action-controllable infinite video generation
753 emerges from autoregressive self-rollout. In CVPR,
754 2026. 3
- 755 [24] Tianwei Yin, Qiang Zhang, Richard Zhang, William T
756 Freeman, Fredo Durand, Eli Shechtman, and Xun
757 Huang. From slow bidirectional to fast autoregressive
758 video diffusion models. In CVPR, 2025. 2
- 759 [25] Dian Zheng, Ziqi Huang, Hongbo Liu, Kai Zou, Yinan
760 He, Fan Zhang, Yuanhan Zhang, Jingwen He, Wei-
761 Shi Zheng, Yu Qiao, and Ziwei Liu. Vbench-2.0: Ad-
762 vancing video generation benchmark suite for intrinsic
763 faithfulness. ArXiv, abs/2503.21755, 2025. 2
- 764 [26] Hongzhou Zhu, Min Zhao, Guande He, Hang Su,
765 Chongxuan Li, and Jun Zhu. Causal forcing: Au-
766 toregressive diffusion distillation done right for high-
767 quality real-time interactive video generation. arXiv
768 preprint arXiv:2602.02214, 2026. 3